

Paper Completion Checklist

Representations of Lie Groups: A Mathematical Dissection of Quantum Spin

9-day sprint — July 20, 2026

Day 1 — $\mathfrak{so}(3)$ & $\mathfrak{su}(2)$ Lie Algebras

- ☐ Define $\mathfrak{so}(3) = T_{I_3}SO(3) = \{\mathbf{R} \in \mathbb{R}^{3,3} \mid \mathbf{R}^T = -\mathbf{R}\}$
- ☐ Define basis $\mathbf{L}_1, \mathbf{L}_2, \mathbf{L}_3$ with explicit matrices
- ☐ Prove $[\mathbf{L}_i, \mathbf{L}_j] = \varepsilon_{ijk} \mathbf{L}_k$ and $\dim \mathfrak{so}(3) = 3$
- ☐ Define $\mathfrak{su}(2) = T_{I_2}SU(2) = \{\mathbf{U} \in \mathbb{C}^{2,2} \mid \mathbf{U}^\dagger = -\mathbf{U}, \text{tr}(\mathbf{U}) = 0\}$
- ☐ Define basis $\mathbf{T}_i = \frac{1}{2i} \sigma_i$ with explicit Pauli matrices
- ☐ Prove $[\mathbf{T}_i, \mathbf{T}_j] = \varepsilon_{ijk} \mathbf{T}_k$ and $\dim \mathfrak{su}(2) = 3$

Day 2 — Double Cover & Lie Algebra Isomorphism

- ☐ Define $\Phi : SU(2) \rightarrow SO(3)$ by $\Phi(\mathbf{U})(\mathbf{X}) = \mathbf{U} \mathbf{X} \mathbf{U}^\dagger$
- ☐ Prove $\Phi(\mathbf{U}) \in SO(3)$ (preserves inner product and determinant)
- ☐ Prove Φ is a smooth group homomorphism
- ☐ Prove Φ is surjective with $\ker \Phi = \{\pm I_2\}$ (2-to-1)
- ☐ Conclude $SU(2)$ is the universal double cover of $SO(3)$
- ☐ Define $\phi : \mathfrak{so}(3) \rightarrow \mathfrak{su}(2)$ by $\phi(\mathbf{L}_i) = \mathbf{T}_i$
- ☐ Prove ϕ is a Lie algebra isomorphism (linear + preserves bracket)

Day 3 — Complexification & Ladder Operators

- ☐ Define complexification: $\mathfrak{g}_{\mathbb{C}} = \mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C}$ for a real Lie algebra $\mathfrak{g}$
- ☐ Define complex representation of a real Lie algebra (extends to $\mathfrak{g}_{\mathbb{C}}$)
- ☐ Prove $\mathfrak{su}(2)_{\mathbb{C}} \cong \mathfrak{sl}(2, \mathbb{C}) = \{\mathbf{A} \in \mathbb{C}^{2,2} \mid \text{tr}(\mathbf{A}) = 0\}$
- ☐ Define Cartan–Weyl basis: $\mathbf{H} = 2i\mathbf{T}_3, \mathbf{E} = i\mathbf{T}_1 - \mathbf{T}_2, \mathbf{F} = i\mathbf{T}_1 + \mathbf{T}_2$
- ☐ Write explicit matrix forms of $\mathbf{H}, \mathbf{E}, \mathbf{F}$
- ☐ Prove commutation relations: $[\mathbf{H}, \mathbf{E}] = 2\mathbf{E}, [\mathbf{H}, \mathbf{F}] = -2\mathbf{F}, [\mathbf{E}, \mathbf{F}] = \mathbf{H}$

Days 4–5 — Classification of Irreducible Representations (the big one)

- ☐ State the theorem: for each $j \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$, $\exists!$ irrep $\pi_j : \mathfrak{su}(2) \rightarrow \mathfrak{gl}(V_j)$ with $\dim V_j = 2j+1$
- ☐ Define weight spaces and prove $\pi_j(\mathbf{H})$ diagonalizes on V_j
- ☐ Ladder Lemma: $\pi_j(\mathbf{E})$ raises weight by $+2$, $\pi_j(\mathbf{F})$ lowers by -2
- ☐ Highest weight argument: pick $\mathbf{v} \neq \mathbf{0}$ with $\pi_j(\mathbf{E})\mathbf{v} = \mathbf{0}$, let λ be its $\mathbf{H}$ -eigenvalue
- ☐ Construct basis $\mathbf{v}_k = \pi_j(\mathbf{F})^k \mathbf{v}$ and prove $\pi_j(\mathbf{E})\mathbf{v}_k = k(\lambda - k + 1)\mathbf{v}_{k-1}$
- ☐ Prove $\lambda = 2j \in \mathbb{N}_0$ and the chain terminates at $\mathbf{v}_{2j+1} = \mathbf{0}$
- ☐ Conclude $\dim V_j = 2j+1$ and weights are $\{-2j, -2j+2, \dots, 2j-2, 2j\}$
- ☐ Normalize to get the standard ladder formulas with square-root factors

- ☐ Prove uniqueness: any irrep of dimension $2j + 1$ is equivalent to π_j
- ☐ Classification table: $j = 0$ scalar, $\frac{1}{2}$ spinor, 1 vector, $\frac{3}{2}, \dots$

Day 6 — Integration to $SU(2)$

- ☐ Prove $SU(2) \cong \mathbb{S}^3$ (homeomorphic to the 3-sphere)
- ☐ Conclude $SU(2)$ is simply connected
- ☐ Theorem: every Lie algebra rep $\pi : \mathfrak{su}(2) \rightarrow \mathfrak{gl}(V)$ integrates uniquely to a Lie group rep $\Pi : SU(2) \rightarrow GL(V)$ via $\Pi(e^X) = e^{\pi(X)}$
- ☐ Apply to π_j to get $\Pi_j : SU(2) \rightarrow GL(V_j)$

Day 7 — Projective Representations & Descent

- ☐ Define projective representation: $\rho(g)\rho(h) = \omega(g, h)\rho(gh)$
- ☐ Bargmann's theorem / physical motivation: in QM, symmetries need only be represented up to phase
- ☐ Prove: every projective representation of $SO(3)$ lifts to an ordinary representation of $SU(2)$
- ☐ Prove descent criterion: $\Pi_j(-I_2) = e^{2\pi \cdot \pi_j(T_3)} = (-1)^{2j} I_{V_j}$
- ☐ Conclude: integer j descends to ordinary $SO(3)$ rep; half-integer j gives projective $SO(3)$ rep (spinor)

Day 8 — Physical Spin Operators

- ☐ Define Hermitian spin observables: $S_k = i\hbar \pi_j(T_k)$
- ☐ Prove angular momentum algebra: $[S_i, S_j] = i\hbar \varepsilon_{ijk} S_k$
- ☐ Prove $S^2 = \hbar^2 j(j+1)I$ and S_z has eigenvalues $\hbar m$ ($m = -j, \dots, j$)
- ☐ Example: $j = \frac{1}{2}$ — Pauli matrices $S_k = \frac{\hbar}{2} \sigma_k$ with explicit matrix forms
- ☐ Spinor sign change: rotation by 2π maps $|\psi\rangle \mapsto -|\psi\rangle$; restored after 4π

Day 9 — Polish & Submit

- ☐ Proof-read all new content for mathematical correctness
- ☐ Ensure consistent notation with earlier sections
- ☐ Add cross-references (`\ref`) throughout new sections
- ☐ Run full compile with `latexmk` — zero errors, zero warnings
- ☐ Update abstract if needed to reflect completed content
- ☐ Final `./publish` to push to GitHub

Feasibility: Yes, 9 days is realistic.

- Days 1–3 are straightforward (definitions + short proofs) — **easy**.
- Days 4–5 (classification theorem) is the hardest — **2 focused sessions**.
- Days 6–8 are moderate (integration, descent, spin operators) — **standard Lie theory**.
- Day 9 is polish — **light work**.
- Total: **~17 new items**, most short. Your writing speed from the original 1,916-line build suggests this is well within range.

